Diabetes and Obesity Prevalence in Zambia

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Introduction

Non-Communicable Diseases (NCDs), also known as chronic diseases, are not mainly caused by acute infection, they tend to be the result of a combination of genetic, physiological, environmental, and behavioral factors. NCDs result in long-term health consequences and often create a need for long-term treatment and care. Type 2 diabetes mellitus (DM) and obesity are classified under this category of diseases.

Obesity or excessive weight gain is identified by health research as the most important and significant risk factor in the development and progression of type 2 DM in all age groups. Zambia has experienced a growth in the prevalence of type 2 DM and obesity in the past three decades.

Overview

This summary focuses on the prevalence rates of Diabetes and Obesity and the relationship between them in Zambia from the year 1990 to 2022. Both show an increase in prevalence rates over the years, with notable variations between genders and ages.

Obesity

Overweight and obesity are defined as abnormal or excessive fat accumulation that presents a risk to health. Obesity is recognized by the World Health Organization (WHO) and major health organizations as a chronic disease and a primary, modifiable risk factor for the development of major NCDs.

The prevalence rates for overweight and obesity are not the same thing. Obesity prevalence rates represent the percentage of adults aged 18+ years with a body mass index (BMI) of 30 kg/m2 or higher and the percentage of children and adolescents with a BMI greater than 2 standard deviations above the median of the population. While overweight prevalence is percentage of adults aged 18+ years with a BMI of 25 kg/m2 or higher and the percentage of children and adolescents with a BMI greater than 1 standard deviation above the population median.

Diabetes

Diabetes is a chronic disease that occurs either when the pancreas does not produce enough insulin or when the body cannot effectively use the insulin it produces. This summary will use diabetes and type 2 diabetes mellitus interchangeably.

The prevalence rates used were age-standardized rates, to eliminate the influence of varying age distributions, and crude rates, which represent the total proportion of the population suffering from diabetes.

Key Findings

- The adult population has a higher prevalence of obese and overweight individuals. The aging process, accompanied by hormonal shifts and a decrease in physical activity, contributes to an increased risk of obesity.
- Females have a higher prevalence of obese and overweight individuals across all age groups except for obese children and adolescents where males have a higher prevalence rate. Pregnancy is a contributing factor. Some women have trouble in losing weight gained during pregnancy, which can lead to long-term obesity or overweight.
- From 1990 the prevalence rate of diabetes in the country. Females over the years have a higher prevalence rate than males. This rise in the prevalence rates adds a strain to the country's health system.
- The treatment coverage which is the percentage of people aged over 30 years with diabetes who are currently using glucose-lowering medication increased steadily from the year 1990 to 2004 and then begun to drop in the year 2005. During the early to mid-2000s, while antiretroviral therapy (ART) was introduced, NCDs like diabetes received relatively little attention or resources from government and international programs compared to infectious diseases like HIV, TB, and malaria.
- Treatment coverage across the country has been below 20%. Traditional healers are an integral part of the health care systems in Zambia. This may indicate the means of treatment the other proportion of diabetic patients are utilizing.
- Over the entire period a greater proportion of males have been receiving treatment compared to females.
- Obesity and diabetes have a strong positive correlation of 0.99. A rise in obesity cases over the years correlated with a rise in diabetes.

Limitations

Not all the available data is nationally representative. Some studies only measured Fasting Plasma Glucose (FPG) or only measured glycated hemoglobin (HbA1c) for the prevalence of diabetes.

Conclusion and Recommendations

The rise in prevalence rates calls for increased attention to Zambia's health financing and labor force in the health sector. They may also result in an economic burden through productivity losses and GDP losses for the country.

More sensitizations should be implemented in the relationship between obesity and diabetes. Individuals can lower their risk of developing diabetes by closely monitoring their BMI and lifestyle to avoid getting overweight or worst case obese.